Pairwise Balanced Designs with Controlled Block Sizes

Theorem 1. *For all sufficiently large integers n, there exists a non-trivial family $\mathcal{A} = \{A_1, A_2, \ldots, A_m\} \subseteq 2^{[n]}$ such that:*

1. (Pairwise balanced) *For all distinct $x, y \in [n]$, there exists exactly one index i such that $\{x, y\} \subseteq A_i$.*
2. (Size control) *There exists a constant $C > 0$ such that for all integers $t \geq 2$,*

$$\#\{i : |A_i| = t\} \leq C\sqrt{n}.$$

Proof. Throughout, define $f(t) := \#\{i : |A_i| = t\}$, $m := \sum_t f(t)$ (total number of blocks), and $\sigma := \sum_t t\, f(t)$ (total number of incidences).

Step 1 (Pair-count identity). Let $\mathcal{A} = \{A_1, \ldots, A_m\}$ be any pairwise balanced family on $[n]$. Consider the set

$$S = \big\{(i, \{x, y\}) : \{x, y\} \subseteq A_i,\ x \neq y\big\}.$$

Counting $|S|$ by block gives $|S| = \sum_{i=1}^m \binom{|A_i|}{2}$. Counting by pair, the pairwise balance condition ensures each of the $\binom{n}{2}$ pairs $\{x, y\} \subseteq [n]$ contributes exactly 1 to $|S|$, giving $|S| = \binom{n}{2}$. Equating:

$$\sum_{i=1}^{m} \binom{|A_i|}{2} = \binom{n}{2}. \tag{1}$$

Partitioning blocks by size, and noting that blocks of size 0 or 1 contribute 0, identity (1) becomes

$$\sum_{t \geq 2} f(t) \binom{t}{2} = \binom{n}{2}. \tag{2}$$

Step 2 (Feasibility of a controlled frequency sequence). Set $T := \lfloor n^{2/3} \rfloor$. Since $T \leq n^{2/3}$,

$$3\binom{T+1}{3} = \frac{(T+1)T(T-1)}{2} \leq \frac{(n^{2/3}+1)n^{2/3}(n^{2/3}-1)}{2} = \frac{n^2 - n^{2/3}}{2} \leq \frac{n^2 - n}{2} = \binom{n}{2}.$$

Define

$$\rho := \binom{n}{2} - 3\binom{T+1}{3} \geq 0.$$

From the expansion $3\binom{T+1}{3} = \frac{n^2}{2} + O(n^{4/3})$, there exists an absolute constant $A > 0$ such that $0 \leq \rho \leq An^{4/3}$. For all sufficiently large n,

$$\binom{T}{2} \geq \frac{(n^{2/3}-1)(n^{2/3}-2)}{2} \geq \frac{n^{4/3}}{4},$$

so setting $K := 4A$ gives $\rho \leq K\binom{T}{2}$.

Define coefficients δ_t via the Euclidean division algorithm as follows. Set $\rho^{(T)} := \rho$. For $t = T, T-1, \ldots, 2$, write

$$\rho^{(t)} = \delta_t \binom{t}{2} + \rho^{(t-1)}, \qquad 0 \leq \rho^{(t-1)} < \binom{t}{2}, \qquad \delta_t = \left\lfloor \frac{\rho^{(t)}}{\binom{t}{2}} \right\rfloor \geq 0.$$

After step $t = 2$, the remainder satisfies $\rho^{(1)} < \binom{2}{2} = 1$, hence $\rho^{(1)} = 0$, and $\sum_{t=2}^{T} \delta_t \binom{t}{2} = \rho$. The coefficient at $t = T$ satisfies $\delta_T \leq K$. For $3 \leq t \leq T - 1$,

$$\delta_t < \frac{\binom{t+1}{2}}{\binom{t}{2}} = \frac{t+1}{t-1} < 2,$$

so $\delta_t \in \{0, 1\}$. For $t = 2$, $\delta_2 \in \{0, 1, 2\}$.

Define the frequency sequence

$$f(t) := \begin{cases} 3 + \delta_t & 2 \leq t \leq T, \\ 0 & t > T. \end{cases}$$

Then $f(t) \geq 3 > 0$ for all $2 \leq t \leq T$, and

$$\sum_{t \geq 2} f(t) \binom{t}{2} = \sum_{t=2}^{T} (3 + \delta_t) \binom{t}{2} = 3 \binom{T+1}{3} + \sum_{t=2}^{T} \delta_t \binom{t}{2} = 3 \binom{T+1}{3} + \rho = \binom{n}{2}.$$

Moreover, $f(T) \leq 3 + K$, $f(t) \leq 4$ for $3 \leq t \leq T - 1$, and $f(2) \leq 5$, so $f(t) = O(1)$ for all t. In particular, $f(t) \leq C_0$ for some absolute constant C_0, hence $f(t) \leq C_0 \leq C_0\sqrt{n}$ for all sufficiently large n.

Step 3 (Existence of PBDs for all sufficiently large n). Let $K_0 = \{2, 3\}$. Then

$$\gcd\left\{ \binom{k}{2} : k \in K_0 \right\} = \gcd(1, 3) = 1, \qquad \gcd\{k - 1 : k \in K_0\} = \gcd(1, 2) = 1.$$

Since both gcds equal 1, the necessary divisibility conditions

$$\binom{n}{2} \equiv 0 \pmod{\gcd\left\{ \binom{k}{2} : k \in K_0 \right\}}, \qquad n - 1 \equiv 0 \pmod{\gcd\{k - 1 : k \in K_0\}}$$

hold for every integer n. By Wilson's theorem on pairwise balanced designs, there exists n_0 such that for all $n \geq n_0$, a pairwise balanced design on $[n]$ with all block sizes in $K_0 = \{2, 3\}$ exists.

Step 4 (Lower bound: some frequency is at least $c\sqrt{n}$). We establish that the size-control bound in the main theorem is sharp up to constants by proving that for any non-trivial PBD on $[n]$, some $f(t)$ must be at least $c\sqrt{n}$.

For each $x \in [n]$, let r_x denote the number of blocks containing x. Summing $\sum_{j : x \in A_j} (|A_j| - 1) = n - 1$ over all x gives

$$\sum_t f(t)\, t(t-1) = n(n-1), \qquad \text{equivalently} \qquad \sum_t t^2 f(t) = n(n-1) + \sigma. \tag{3}$$

Since any two distinct blocks share at most one point (as the family is pairwise balanced),

$$\sum_x r_x^2 = \sigma + \sum_{i \neq j} |A_i \cap A_j| \leq \sigma + m(m-1).$$

By the Cauchy–Schwarz inequality,

$$\sigma^2 = \left(\sum_x r_x \right)^2 \leq n \sum_x r_x^2 \leq n\sigma + nm^2,$$

so $(\sigma - n/2)^2 \leq nm^2 + n^2/4$, giving

$$\sigma \leq n + m\sqrt{n}. \tag{4}$$

From (3) and the Cauchy–Schwarz inequality $\sigma^2 \leq m \sum_t t^2 f(t) = m(n(n-1) + \sigma)$:

$$mn(n-1) \leq \sigma^2 \leq (n + m\sqrt{n})^2 = n^2 + 2nm\sqrt{n} + m^2 n.$$

Dividing by n and rearranging,

$$m\big(n-1-2\sqrt{n}-m\big) \le n.$$

If $m \le n/3$, then for large n the factor $n-1-2\sqrt{n}-m \ge n/4$, giving $m \le 4$. But $m \le 4$ blocks cannot cover all $\binom{n}{2}$ pairs for large n, a contradiction. Hence

$$m \ge \frac{n}{3} \quad \text{for all sufficiently large } n. \tag{5}$$

From (4) and (5), the weighted mean square block size satisfies

$$\mathbb{E}[t^2] := \frac{1}{m}\sum_t t^2 f(t) \le \frac{n^2}{m} + 2\sqrt{n} \le 3n + 2\sqrt{n} \le 4n,$$

so the weighted variance satisfies $\mathrm{Var}(t) \le \mathbb{E}[t^2] \le 4n$. By Chebyshev's inequality applied to the distribution $\mathbb{P}(t=k) = f(k)/m$,

$$\mathbb{P}\big(|t-\mathbb{E}[t]| > 4\sqrt{n}\big) \le \frac{4n}{16n} = \frac{1}{4}.$$

Hence at least $\frac{3}{4}m \ge \frac{n}{4}$ blocks have size in the interval $I = [\mathbb{E}[t]-4\sqrt{n}, \mathbb{E}[t]+4\sqrt{n}]$, which contains at most $8\sqrt{n}+1 \le 9\sqrt{n}$ integer values. By the pigeonhole principle, some t^* satisfies

$$f(t^*) \ge \frac{n/4}{9\sqrt{n}} = \frac{\sqrt{n}}{36}.$$

Setting $c = 1/36$ gives $\max_t f(t) \ge c\sqrt{n}$.

Step 5 (Weak frequency control: $\max_t f(t) \le n^\alpha$ for some $\alpha < 1$). Let q be a prime power and consider the projective plane $\mathrm{PG}(2,q)$, which has $M = q^2+q+1$ points and lines, each line of size $q+1$, with every pair of points in exactly one line. Include each point independently with probability $p = n/M$ for a target size $n = \Theta(q^2)$. Let S be the resulting random subset, and define

$$\mathcal{A} = \{L \cap S : L \text{ is a line},\ |L \cap S| \ge 2\}.$$

For any distinct $x, y \in S$, the unique line L of $\mathrm{PG}(2,q)$ through them satisfies $|L\cap S| \ge 2$, so $L \cap S \in \mathcal{A}$, and no other block covers $\{x,y\}$. Hence $\mathcal{A}$ is a pairwise balanced design on S.

For each line L, $X_L := |L \cap S| \sim \mathrm{Binomial}(q+1, p)$, and the local central limit theorem gives

$$\max_t \mathbb{P}(X_L = t) \le \frac{C}{\sqrt{(q+1)p(1-p)}}.$$

Since $(q+1)p = \Theta(n/q)$ and $q \asymp \sqrt{n}$,

$$\mathbb{E}[f(t)] = M\,\mathbb{P}(X_L = t) \le CM\sqrt{\frac{q}{n}} \le Cq^2 \cdot \frac{q^{1/2}}{\sqrt{n}} = Cq^{3/2} = O(n^{3/4}).$$

Changing one point changes at most $q+1 = O(q)$ lines, hence changes $f(t)$ by $O(q)$. With $M = \Theta(q^2)$ independent variables each of Lipschitz constant $O(q)$, McDiarmid's inequality gives, for $\lambda = n^{0.8}$,

$$\mathbb{P}\big(|f(t)-\mathbb{E}[f(t)]| > n^{0.8}\big) \le 2\exp\big(-cn^{0.6}\big).$$

Since there are $O(q) = O(\sqrt{n})$ possible values of t, a union bound gives

$$\mathbb{P}\big(\exists\, t : |f(t)-\mathbb{E}[f(t)]| > n^{0.8}\big) \le \sqrt{n}\,e^{-cn^{0.6}} \to 0.$$

Since $|S| \sim \mathrm{Binomial}(M,p)$ with mean n and variance $O(n)$, we have $|S| = n + O(\sqrt{n})$ with high probability. Adjusting $O(\sqrt{n})$ points changes each $f(t)$ by $O(q\sqrt{n}) = O(n^{3/4})$. Hence there exists a realisation S with $|S| = n$ such that for all t,

$$f(t) \le Cn^{3/4} + n^{0.8} \le C'n^{0.8}.$$

Relabelling S as $[n]$ gives a PBD on $[n]$ with $\max_t f(t) \le C' n^{\alpha}$ for $\alpha = 0.8 < 1$.

Step 6 (Target frequency bound: $f(t) \le C\sqrt{n}$ for all t). By Step 2, there exist nonnegative integers $f(t)$ satisfying

$$\sum_{t \ge 2} f(t) \binom{t}{2} = \binom{n}{2}, \qquad f(t) \le C_0 \text{ for all } t,$$

with block sizes supported on $\{2, \dots, T\}$. In particular,

$$\gcd\left\{ \binom{t}{2} : f(t) > 0 \right\} \le \gcd\left(\binom{2}{2}, \binom{3}{2} \right) = \gcd(1, 3) = 1,$$

and

$$\gcd\{t - 1 : f(t) > 0\} \le \gcd(1, 2) = 1.$$

Both gcds equal 1, so the necessary divisibility conditions of Wilson's theorem are satisfied for all n. By Wilson's theorem, for all sufficiently large n, there exists a pairwise balanced design on $[n]$ realising the frequency sequence $f(t)$. Since $f(t) \le C_0$ for all t, and $C_0 \le C_0\sqrt{n}$ for all $n \ge 1$, the resulting PBD satisfies

$$\max_t f(t) \le C_0 \le C\sqrt{n}$$

for $C = C_0$.

Step 7 (Realization and conclusion). The necessary and sufficient conditions for a sequence of nonneg integers $f(t)$ to be realised by a PBD on $[n]$ are:

$$\sum_{t \ge 2} f(t) \binom{t}{2} = \binom{n}{2}, \qquad \gcd\left\{ \binom{t}{2} : f(t) > 0 \right\} \mid \binom{n}{2}, \qquad \gcd\{t - 1 : f(t) > 0\} \bigm| (n - 1).$$

The frequency sequence constructed in Step 2 satisfies all three conditions for every n, as both gcds equal 1. Wilson's theorem then guarantees the existence of a PBD on $[n]$ with exactly $f(t)$ blocks of size t for all sufficiently large n.

Combining Steps 2 and 6: for all sufficiently large n, there exists a pairwise balanced design $\mathcal{A}$ on $[n]$ with

$$\sum_{t \ge 2} f(t) \binom{t}{2} = \binom{n}{2} \qquad \text{and} \qquad f(t) \le C\sqrt{n} \text{ for all } t \ge 2,$$

where $C = C_0$ is the absolute constant from Step 2. This is the conclusion of the main theorem.

$\square$